Sharp Mixed-Potential Auxiliaries and Quantitative Energy Transfer

A Scoped Analysis of Universal Optimality of the Triangular Lattice

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Technical Report Assisted by AI Models
www.integralresearch.net
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Technical Report TR-Ram-OpenAI-722-M182

October 8, 2026

Abstract

The September 23 source constructs a sharp Gaussian family through a fixed infinite interpolation operator, then integrates energy inequalities rather than auxiliary functions. We prove a conditional strengthening tailored to that construction: a canonical measurable family can be integrated to an actual sharp auxiliary for every positive Gaussian mixture satisfying an explicit inverse-moment condition. Integrability is characterized exactly; Schwartz regularity requires all inverse moments, not merely integrability of the potential away from the origin. The proof uses the source's compact complex spectral realization and its bounded infinite inverse, not finite matrix invertibility alone. We also establish an exponentially weighted inverse domain, explicit residual allowances, and finite-disk energy bounds without separation or local occupancy assumptions. Source certificate reruns and independent rational checks are distinguished from proof of the analytic reductions. Renormalized screening, universal interpolation uniqueness, and the global optimality claim are not independently certified. Contribution assessment: C3/5, a source-specific mixed-auxiliary construction with exact admissibility thresholds and quantitative transfer, conditional on the stated Gaussian certificate and analytic inputs.

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1 Problem, scope, and dependency map

Let $b = \sqrt{3}/2$, and put

$$A = b^{-1/2}\{j(1,0) + \ell(1/2, b) : j, \ell \in \mathbb{Z}\}.$$

Its covolume is one; its dual is a rotation of A. Our Fourier convention is $\widehat{f}(\xi) = \int_{\mathbb{R}^2} f(x)e^{-2\pi i x\cdot\xi}\,dx$. Write $G_\alpha(x) = e^{-\pi\alpha|x|^2}$, so $\widehat{G}_\alpha = \alpha^{-1}G_{1/\alpha}$. For a locally finite simple configuration $\mathcal{C}$, set $N_R = \#(\mathcal{C} \cap B_R)$ and use ordered pairs:

$$E_g(\mathcal{C}) = \liminf_{R\to\infty} \frac{1}{N_R} \sum_{\substack{x,y\in\mathcal{C}\cap B_R \\ x\neq y}} g(|x-y|^2).$$

The source [1] claims $E_g(\mathcal{C}) \geq E_g(A)$ for every nonnegative completely monotone g and every such configuration with $N_R/(\pi R^2) \to 1$, allowing infinite energies and arbitrarily dense local clusters. We attribute this theorem to the source. Neither a numerical rerun nor the local deductions below is an independent proof audit of that entire assertion.

The complete cached scientific text was literally read: all ten section files, 5,060 lines; the main file, macros, bibliography, and public supporting texts add 502 lines. Reading a cited theorem is not reading its proof in the cited work. The accompanying ledger records the exact display intervals, before/after hashes, and the two private original passages read locally.

Source component	Mathematical obligation and treatment here
01, introduction	All centered-density competitors; extended energies; no all-Schwartz uniqueness or mixed-potential auxiliary asserted.
02, cardinal Fourier	Sine cardinal functions, finite complex spectral measures, exact Fourier realization and kernel bounds. Used explicitly.
03–04, finite data and inversion	Fejer quadrature, 168-coordinate block per list, paired 336-block, three residuals, four infinite blocks and Schur complement. Numerical bounds retained as source inputs.
05, global signs	Deleted jets before division; Bernstein positivity on 91 half-gaps, then analytic tails. No sampling-only substitution.
06–07, energy and mixtures	Positive-definite Cauchy–Schwarz, Hausdorff mixtures, Fatou on arbitrary radius subsequences, shifted limit. Quantitative and functional extensions are proved here.
08, renormalized energy	Heat representation and ordered cutoff/box limits. The 1,059-line screening appendix inside arithmetic verification is read but not independently reproved or imported into our local theorems.
Arithmetic and floor appendices	Scalar errors; separate conditional floor-rounded procedure. The latter is not executed by the supplied checkers.

The source's mixture proof retains the endpoint corresponding to a constant potential as infinite energy; removes the spurious zero moment atom; uses Fatou along an arbitrary sequence of radii; and takes the final shifted monotone limit only on the lattice sum. These are useful valid distinctions, not gaps to be repaired by assuming uniform separation. Its renormalized argument additionally controls close-pair defects before screening and removes the cutoff after the box limit. Our ordinary-energy results do not replace any of those renormalized obligations.

2 The actual interpolation model

Set $h = 2/5$, $B = 4/3$, $s = b|x|^2$ and

$$\mathcal{N} = (12\mathbb{Z} + \{0,1,3,4,7,9\}) \cap \mathbb{Z}_{>0}, \qquad P(s) = \prod_{a\in\{0,1,3,4,7,9\}} \left(2\sin\frac{\pi(s-a)}{12}\right)^2.$$

These nodes strictly contain the lattice shell indices; for example 24 is retained although it is not a lattice shell. A full list $a = (C, (c_n, d_n)_{n\in\mathcal{N}\cup\{0\}})$ has norm $|C| + \sum_n(|c_n| + |d_n|)$ and defines

$$p_a(s) = P(s)\left[C + \sum_n \left(\frac{c_n}{(s-n)^2} + \frac{d_n}{s-n}\right)\right]. \tag{1}$$

Removable values are understood. With $P(n+u) = Q_n u^2(1 + D_n u + O(u^2))$, one has $p_a(n) = Q_n c_n$ and $p_a'(n) = Q_n(d_n + D_n c_n)$. The source establishes $Q_n > 1.75$, $|D_n| < 2.12$, and the representation

$$p_a(s) = \int_{-1}^{1} e^{\mathrm{i}\pi t s}\, d\mu_a(t), \qquad \|\mu_a\|_{\mathrm{TV}} \le 25|C| + 36\sum_n(|c_n| + |d_n|). \tag{2}$$

The integral, not a real-part formula on a folded interval, defines the entire extension. Real lists give conjugate-symmetric measures. Let

$$z(t) = -\frac{B}{t + \mathrm{i}h} - \mathrm{i}h, \quad \lambda(t) = \frac{\mathrm{i}}{b(t+\mathrm{i}h)}, \quad Kp(s) = \int \lambda(t) e^{\mathrm{i}\pi z(t)s}\, d\mu_a(t).$$

The Fourier transform of $e^{-\pi h b|x|^2} p_a(b|x|^2)$ is $e^{-\pi h b|\xi|^2} K p_a(b|\xi|^2)$. Indeed each integrand is a Gaussian with parameter $b(h - \mathrm{i}t)$ of strictly positive real part, and its inverse parameter has real part $h/[b(h^2+t^2)] > 0$. Compactness of $[-1,1]$ and (2) justify all integrals and every Schwartz seminorm estimate. The identities $z(z(t)) = t$ and $\lambda(t)\lambda(z(t)) = 1$ do not justify composing folded real contours.

Let $E = \ell^1(\mathcal{N}; \mathbb{R}^2)$ with the sum norm, let $Ju = (u(n)/Q_n, (u'(n) - D_n u(n))/Q_n)_n$, and let $S = -JKp$ on positive coordinates. The three extras are $e = (c_0, d_0, C)$, with operator S_e. The paired system on $E \oplus E$ is

$$L\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} I & -S \\ -S & I \end{pmatrix}\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} JT_k + S_e e_2(k) \\ S_e e_1(k) \end{pmatrix}, \quad T_k(s) = k^{-1}e^{-k(s-1)}. \tag{3}$$

In the order (c_0, d_0, C) for each list, the extras are

$$\begin{aligned}(e_1, e_2)(k) = {} & (.567, .250, -.0074, 0, -.209, .0096) \\ & + k^{-1}(-.976, .408, .0115, 1, .938, -.017).\end{aligned}$$

All decimals in this report are exact terminating rationals.

Declared source input G. The exact spectral realization, bounded invertibility of L on E^2, and the source's global signs/contact conclusions are assumed when marked **G**. For every real $k \ge 2.36$, its exact full paired list has norm less than 5, and $H_1 = p_{a_1} + Kp_{a_2}$, $H_2 = p_{a_2} + Kp_{a_1}$ obey $H_1 \le T_k$, $H_2 \ge 0$, with the prescribed positive-node jets. The analytic dependence and measure classifications below are deductions, not additional numerical hypotheses hidden inside G.

3 Infinite inversion, weights, and residuals

The private original TR147 passage read for comparison specifies a Banach space, actual operator realization, and two-sided bounded inversion before using an infinite triangular formula. It also gives the useful warning $x_j - 2x_{j+1}$ on $\ell^2(\mathbb{N}_0)$: every leading finite matrix has determinant one, while $(2^{-j})_j$ is an infinite kernel vector. The original volume16 clearance chapter proves the Bernstein coefficient conversion and subdivision criterion, with remainder control separate. These classical arguments are credited as comparison tools; neither private passage proves the source's interpolation or positivity certificate.

For the source split $F = \mathcal{N} \cap [1, 168]$, $T = \mathcal{N} \setminus F$, its analytic and finite inputs are

$$\left\|L_{FF}^{-1}\right\| < 40, \quad \|L_{FT}\| < 4000, \quad \|L_{TF}\|, \ \|L_{TT} - I\| < \delta = 1.1 \cdot 10^{-14}. \tag{4}$$

The bound 40 is *not* the norm of the full inverse. It comes from the finite quadrature matrix D: $\|D^{64}\| < .004$, $\left\|\sum_{j<512} D^j\right\| + \left\|\sum_{j<512}(-D)^j\right\| < 36$, followed by the exact-versus-quadrature perturbation below $2 \cdot 10^{-27}$. The tail Schur complement is within $\delta(1 + 40 \cdot 4000) = 1.760011 \cdot 10^{-9}$ of identity.

Proposition 3.1 (Two-source residual allowance)**.** *Under (4), an approximate solution with residual norms $r_F, r_T \geq 0$ has exact correction bounded by*

$$u_T \leq \frac{r_T + 40\delta r_F}{D_0}, \qquad u_F \leq 40 r_F + 160000 u_T, \qquad D_0 = 1 - 160001\delta > 0. \tag{5}$$

In particular $\|L^{-1}\| < 160002$ in the paired sum norm.

Proof. Eliminate the finite coordinates and invert the tail Schur complement by its norm-convergent Neumann series. This gives the first inequality; insert it into the finite equation for the second. The coefficients of r_F, r_T in their sum are $40 + 160001(40\delta)/D_0$ and $160001/D_0$, respectively, both below 160002. This proves a two-sided bounded inverse, not just a formal recursive solution. □

For the source reference lists the three finite errors are omitted target jets $< 8 \cdot 10^{-12}$, quadrature $< 10^{-24}$, and finite-series residual $< 1.4 \cdot 10^{-15}$. They give $r_F < 10^{-11}$; the separate analytic tail gives $r_T < 8.3 \cdot 10^{-14}$. Formula (5) yields $u_F + u_T < 1.3681 \cdot 10^{-8} < 2 \cdot 10^{-8}$. Reducing only the finite residual cannot eliminate the amplified tail term.

For a positive node m, delete the zeroth and first jets before dividing:

$$D_{m,2} v(s) = \frac{v(s) - v(m) - (s-m)v'(m)}{(s-m)^2} = \int_0^1 (1-t) v''(m + t(s-m))\, dt.$$

The source derivative estimates give a correction cost at most $486\pi^2(u_F + u_T)$ for this quotient. Thus the finite-half-gap certificate alone survives whenever

$$40 r_F + \frac{160001(r_T + 40\delta r_F)}{D_0} < \frac{.000998}{486\pi^2}. \tag{6}$$

Here $.000998 = .001 - 10^{-6} - 10^{-6}$ reserves the separate polynomial and target errors. This is an explicit admissible residual region, not a new global sign theorem: the rational tail margin, high-list bound, and exact contact jets must still hold. A raw pointwise error divided by $(s-m)^2$ would be invalid.

3.1 An exponentially weighted domain

The exact minimum of the kernel's decay exponent, rather than its rounded lower bound .187, is

$$q_* := \min_{|t|\le 1} \pi \operatorname{Im} z(t) = \pi h \left(\frac{B}{1+h^2} - 1 \right) = \frac{26\pi}{435}. \tag{7}$$

For $0 \le \sigma < q_*$ let E_σ have norm $\sum_{n\in\mathcal{N}} e^{\sigma n}(|c_n| + |d_n|)$, and use the paired sum norm. Define

$$A(t) = \frac{t + t^3 + t^4 + t^7 + t^9 + t^{12}}{1 - t^{12}}, \qquad C_\sigma = 792 A(e^{-(q_* - \sigma)}).$$

Theorem 3.2 (Weighted infinite inverse)**.** *Assume the source's spectral realization and bounded inverse of L on E^2, with norm M. Then L restricts to a boundedly invertible operator on E^2_σ for every $0 \le \sigma < q_*$, and*

$$\left\| L^{-1}\psi \right\|_\sigma \le \|\psi\|_\sigma + C_\sigma M \|\psi\|_0 \le (1 + C_\sigma M) \|\psi\|_\sigma . \tag{8}$$

If $a = L^{-1}\psi$, truncation at node index r costs at most $e^{-\sigma r} \|a\|_\sigma$ in the unweighted norm.

Proof. Since $|\lambda| < 3$, $|z| < 3$, and the normalized value-plus-derivative factor is less than 22, each output node of S is bounded by $22 \|\mu_a\|_{\mathrm{TV}} e^{-q_* n} \le 792 \|a\|_0 e^{-q_* n}$. Summing only the actual periodic nodes, not all integers, gives $\|Sa\|_\sigma \le C_\sigma \|a\|_0$ because $\sum_{n\in\mathcal{N}} t^n = A(t)$. Thus $S : E \to E_\sigma$ is bounded. Solve first in E^2. Its equation says $a = \psi + (Sa_2, Sa_1)$; hence $a \in E^2_\sigma$ and (8) follows. Boundedness of L on the weighted space follows from the same estimate, and uniqueness follows from its inclusion in E^2. The tail inequality uses $e^{-\sigma n} \le e^{-\sigma r}$ for $n \ge r$. □

This bootstraps a *proved infinite inverse*; it does not assume that finite inverse norms remain bounded after reweighting. The constants grow as $\sigma \uparrow q_*$. We make no assertion that the endpoint space is impossible: endpoint cancellation is not measured by this absolute bound.

4 Parameter regularity of the exact family

Proposition 4.1 (Analytic selection and large-parameter bound)**.** *Under G, the solution lists of (3) extend holomorphically to $\{k \in \mathbb{C} : \operatorname{Re} k > 0\}$, as vectors in the complexification of E^2. Their Schwartz realizations are holomorphic in every seminorm. For real $\alpha \ge 1$, put*

$$k = \pi(\alpha/b - h), \qquad f_\alpha(x) = k e^{-k} e^{-\pi h b |x|^2} H_{1,k}(b|x|^2). \tag{9}$$

For every Schwartz seminorm p there is a constant C_p, independent of $\alpha \ge 1$, such that, with $c = \pi/b$,

$$p(f_\alpha) + p(\widehat{f_\alpha}) \le C_p \alpha e^{-c\alpha}. \tag{10}$$

Proof. On a compact subset of $\operatorname{Re} k > 0$, derivatives of $k^{-1}e^{-k(n-1)}$ and $-e^{-k(n-1)}$ are bounded by a fixed polynomial in n times $e^{-\varepsilon(n-1)}$. These bounds are summable over $\mathcal{N}$. Since Q_n^{-1}, D_n are bounded, JT_k is holomorphic as an $E_{\mathbb{C}}$-valued function. The extras are affine in $1/k$ and S_e is bounded. The fixed bounded inverse L^{-1} therefore gives the asserted analytic selection. This remains true in E_σ on the smaller domain $\operatorname{Re} k > \sigma$, using Theorem 3.2.

For completeness, the spectral realization is a continuous linear map from full lists to $\mathcal{S}(\mathbb{R}^2)$. Each spatial derivative of $e^{-\pi\gamma|x|^2}$ is a polynomial in x and γ times that Gaussian. The two parameter sets $\gamma = b(h - it)$ and γ^{-1}, $|t| \le 1$, are compact in the right half-plane. Every polynomially weighted

derivative is uniformly bounded there. Integrating against the total variation in (2) proves continuity, including for the Fourier image. Complexifying the real-linear realization avoids any erroneous use of a real-part formula as holomorphic. For real $k \geq 2.36$ the source's norm bound 5 makes every seminorm uniform. Finally $ke^{-k} \leq ce^{\pi h}\alpha e^{-c\alpha}$, which gives (10). □

For $0 < \alpha < 1$ define, exactly as in the source's duality argument,

$$f_\alpha = G_\alpha - \alpha^{-1}\widehat{f}_{1/\alpha}, \qquad \widehat{f}_\alpha = \alpha^{-1}(G_{1/\alpha} - f_{1/\alpha}). \tag{11}$$

Under G these are real radial Schwartz functions with $f_\alpha \leq G_\alpha$, $\widehat{f}_\alpha \geq 0$, $f_\alpha(v) = G_\alpha(v)$ for $v \in A \setminus \{0\}$, and $\widehat{f}_\alpha(w) = 0$ for $w \in A^* \setminus \{0\}$. The family is Borel measurable in the Schwartz topology: it is analytic on each side of 1, with its value at 1 prescribed by (9). We do not assume the two formulas have the same one-sided limit at 1. That possible mismatch does not obstruct a measure with an atom at 1. Poisson summation for each individual Schwartz function gives

$$\widehat{f}_\alpha(0) - f_\alpha(0) = \sum_{v \in A \setminus \{0\}} e^{-\pi\alpha|v|^2}. \tag{12}$$

5 An actual mixed-potential auxiliary

The source introduction explicitly does not assert an auxiliary function for every completely monotone potential. The following construction supplies one under precise, checkable measure assumptions. Gaussian subordination itself is classical [4]; the increment is the applicability of the source's *specific* exact family, including the behavior at both ends of the Gaussian parameter and the sharp admissibility boundary.

Let ν be a positive Borel measure on $(0, \infty)$ satisfying

$$g(t) = \int_{(0,\infty)} e^{-\pi\alpha t}\, d\nu(\alpha) < \infty \quad \text{for every } t > 0. \tag{13}$$

This implies local finiteness and $\nu((0,1)) < \infty$. An atom at parameter zero, representing a constant potential, is excluded here, not from the source's extended-energy theorem. Write $I_j = \int_{(0,1)} \alpha^{-j}\, d\nu(\alpha)$.

Theorem 5.1 (Exact admissibility classification)**.** *Assume G and (13). The pointwise integral $F_\nu(x) = \int f_\alpha(x)\, d\nu(\alpha)$ exists absolutely for every x, including $x = 0$, and is real, radial, smooth and bounded. More precisely,*

$$F_\nu(x) = g_{<1}(|x|^2) + H_\nu(x), \qquad g_{<1}(t) = \int_{(0,1)} e^{-\pi\alpha t}\, d\nu(\alpha), \quad H_\nu \in \mathcal{S}(\mathbb{R}^2). \tag{14}$$

Its regularity thresholds are exact:

$$F_\nu \in L^1(\mathbb{R}^2) \iff I_1 < \infty, \tag{15}$$

$$F_\nu \in \mathcal{S}(\mathbb{R}^2) \iff I_j < \infty \text{ for every integer } j \geq 1. \tag{16}$$

If $I_1 < \infty$, then $F_\nu, \widehat{F}_\nu \in C_0 \cap L^1$, $\widehat{F}_\nu \geq 0$, and

$$F_\nu(x) \leq g(|x|^2) \quad (x \neq 0), \qquad F_\nu(v) = g(|v|^2) \quad (v \in A \setminus \{0\}), \tag{17}$$

$$\widehat{F}_\nu(w) = 0 \quad (w \in A^* \setminus \{0\}), \qquad \widehat{F}_\nu(0) - F_\nu(0) = \sum_{v \neq 0} g(|v|^2) < \infty. \tag{18}$$

All derivatives of $\widehat{F}_\nu$ decay faster than powers away from the origin. If also $g(t) = O(t^{-1-\varepsilon})$ as $t \to \infty$, then $F_\nu(x) = O(|x|^{-2-2\varepsilon})$ and the Fourier transform decays faster than powers: this gives the usual pointwise-decaying admissible pair.

Proof. First note that for each $d > 0$ and integer $j \geq 0$, $\int_1^\infty \alpha^j e^{-d\alpha}\, d\nu < \infty$: $\alpha^j e^{-d\alpha} \leq C_{d,j} e^{-d\alpha/2}$ and use (13). The large-parameter bound (10) makes

$$H_\nu = \int_{[1,\infty)} f_\alpha \, d\nu - \int_{(0,1)} \alpha^{-1} \widehat{f}_{1/\alpha} \, d\nu$$

converge in every Schwartz seminorm. For the second integral the seminorm is at most $C_p \alpha^{-2} e^{-c/\alpha}$, a bounded function on $(0, 1)$. Integrals over compact parameter intervals are defined by the continuous family; truncation at both ends is Cauchy in every seminorm, hence converges in the complete Schwartz space. This also supplies a concrete meaning to the vector-valued integral without assuming a finite total mass for ν.

The remaining low-parameter term is bounded by $\nu((0, 1))$. All its spatial derivatives can be taken under the integral, because the Gaussian derivative bounds are uniform for $0 < \alpha < 1$ on compact x-sets (indeed uniformly in x for each unweighted derivative). This proves (14) and smooth boundedness. Since $\int_{\mathbb{R}^2} G_\alpha = \alpha^{-1}$, Tonelli gives $\|g_{<1}(|\cdot|^2)\|_1 = I_1$. Adding the integrable H_ν cannot change whether this nonnegative function is integrable. This proves both directions of (15).

If all inverse moments are finite, polynomially weighted Gaussian derivatives are dominated by some $C\alpha^{-j}$ for $0 < \alpha < 1$. Differentiation and integration prove Schwartz regularity. Conversely, if F_ν is Schwartz, so is its nonnegative difference $g_{<1}(|\cdot|^2) = F_\nu - H_\nu$. Polar integration and Tonelli give the exact identities

$$\int_{\mathbb{R}^2} |x|^{2m} g_{<1}(|x|^2)\, dx = \frac{m!}{\pi^m} I_{m+1} \quad (m = 0, 1, 2, \ldots). \tag{19}$$

Their left sides are finite, proving necessity in (16).

Assume $I_1 < \infty$. The family is integrable in L^1 by (11), since its Gaussian term costs α^{-1} and the correction is exponentially small. Its Fourier family is also integrable in L^1: for $\alpha < 1$ the Gaussian term $\alpha^{-1} G_{1/\alpha}$ has L^1 norm 1; its correction is bounded by $C\alpha^{-2} e^{-c/\alpha}$. The large end again uses (10). Fubini proves $\widehat{F}_\nu = \int \widehat{f}_\alpha \, d\nu$. The same estimates in sup norm, using I_1 for the Fourier Gaussian, give continuous pointwise representatives, so positivity and contacts hold at every stated point, not merely almost everywhere. The C_0 assertions follow from L^1 Fourier transformation on both sides.

The signs and direct contacts follow by integrating the exact Gaussian statements. Integrate (12); both zero-value integrals are finite by the preceding bounds, and Tonelli treats its nonnegative series. This proves (18) without applying Poisson summation to a possibly non-Schwartz F_ν.

Finally, for $|\xi| \geq 1$, every derivative of $\int_0^1 \alpha^{-1} e^{-\pi|\xi|^2/\alpha}\, d\nu$ has Gaussian decay: any power of α^{-1} or $|\xi|$ is absorbed by half of the exponential, uniformly on this domain. The Fourier transform of H_ν is Schwartz. Also $0 \leq g_{<1}(t) \leq g(t)$, so (14) proves the last spatial decay assertion. Possible Fourier singularities in higher derivatives at zero are not ruled out by I_1 alone. □

5.1 Examples that separate the domains

For $p > 0$, the potential $g(t) = t^{-p}$ has $d\nu(\alpha) = \pi^p \alpha^{p-1}\, d\alpha/\Gamma(p)$. Thus $I_1 < \infty$ exactly when $p > 1$, and $I_j < \infty$ exactly when $p > j$. For every finite $p > 1$ the constructed auxiliary is integrable and decays as $O(|x|^{-2p})$, but is *not* Schwartz. Its value at zero remains finite despite $g(0+) = \infty$. For

$0 < p \le 1$ this construction has a nonintegrable positive tail; it is not an ordinary sharp integrable LP certificate. No renormalized conclusion follows from subtracting infinities.

For $\tau > 0$, $g(t) = e^{-\pi\tau t} t^{-p}$ has the translated measure $\pi^p(\alpha - \tau)^{p-1}\mathbf{1}_{\alpha>\tau}\, d\alpha/\Gamma(p)$. Its support is bounded away from zero, so all I_j are finite. The auxiliary is Schwartz for every $p > 0$, although the potential is still singular at zero. Positive parameter measures supported on a compact subset of $(0, \infty)$, explicitly bounded away from zero, give another immediate Schwartz class. These examples show why the small parameter end, not the singularity of the potential at spatial zero, controls admissibility of this particular source construction.

6 Finite-window transfer without occupancy bounds

This section is independent of G until it is applied to F_ν. It quantifies the source's asymptotic positive-definite argument. For a real even continuous f with $f, \widehat{f} \in L^1$, $\widehat{f} \ge 0$ and $a = \widehat{f}(0) > 0$, set

$$T_f(s) = \int_{|z|\ge s} |f(z)|\, dz, \qquad B_f(L) = \int_{\mathbb{R}^2} \min\left(1, \frac{2|z|}{\pi L}\right) |f(z)|\, dz.$$

These are finite even when $\int |z||f(z)|\, dz = \infty$.

Theorem 6.1 (Observed-count disk bound)**.** *Let $x_1, \dots, x_N$ be distinct points in B_R, with $N \ge 1$, and let $\Phi(x) \ge f(x)$ for $x \ne 0$. For every $s > 0$, with $L = R + s$,*

$$\frac{1}{N}\sum_{i\ne j} \Phi(x_i - x_j) \ge \frac{N}{\pi L^2}\frac{(a - T_f(s))_+^2}{a + B_f(L)} - f(0). \tag{20}$$

If $M_1 = \int |z||f(z)|\, dz < \infty$, one may replace $B_f(L)$ in the denominator by $2M_1/(\pi L)$. No separation, density, or local-count assumption is needed for this finite statement.

Proof. For finite measures define the positive semidefinite form

$$Q(\mu, \lambda) = \iint f(x - y)\, d\mu(x)\, d\overline{\lambda}(y) = \int \widehat{f}(\xi)\widehat{\mu}(\xi)\overline{\widehat{\lambda}(\xi)}\, d\xi,$$

where for the real measures used below the choice of which argument is conjugated is immaterial. Fourier inversion and $\widehat{f} \in L^1$ justify the equality; Cauchy–Schwarz applies even if the form is degenerate. Take $\mu = \sum_i \delta_{x_i}$ and $\lambda = \mathbf{1}_{B_L}\, dx$. Because $B_s(x_i) \subset B_L$, $Q(\mu, \lambda) \ge N(a - T_f(s))$.

For a translation z of length d, slice a disk of radius L parallel to z. Each interval slice loses at most d of overlap and the transverse range has length $2L$. Hence

$$0 \le \pi L^2 - |B_L \cap (B_L - z)| \le \min(\pi L^2, 2Ld).$$

It follows, retaining the sign of $\int f = a$ but bounding the loss by absolute values, that

$$Q(\lambda, \lambda) = \int f(z)|B_L \cap (B_L - z)|\, dz \le \pi L^2(a + B_f(L)).$$

If $a > T_f(s)$, Cauchy–Schwarz and the positive lower cross term give

$$Q(\mu, \mu) \ge \frac{N^2(a - T_f(s))^2}{\pi L^2(a + B_f(L))}.$$

If not, the same assertion with the positive part follows from $Q(\mu, \mu) \ge 0$. Finally subtract the N diagonal terms $f(0)$, divide by N, and use $\Phi \ge f$ off the diagonal. The moment variant follows directly from $B_f(L) \le 2M_1/(\pi L)$. □

Corollary 6.2 (An explicit density and padding error)**.** *Suppose $M_1 < \infty$ and $T_f(s) \le C_m s^{-m}$ for $s \ge 1$, with $m > 0$. For $R \ge 1$, $s = R^{1/(m+1)}$ and $C_m R^{-m/(m+1)} \le a$, let $\rho_R = N/(\pi R^2)$. Then*

$$\frac{1}{N}\sum_{i\neq j}\Phi(x_i - x_j) \ge a - f(0) - a|\rho_R - 1| \\ - \rho_R\left[(2a + 2C_m)R^{-m/(m+1)} + \frac{2M_1}{\pi R}\right]. \tag{21}$$

For a centered-density-one configuration and any f satisfying the theorem, without the moment assumption, the asymptotic lower bound is $E_\Phi \ge a - f(0)$.

Proof. Put $\lambda = s/R$, $\tau = T_f(s)/a$, $\omega = B_f(L)/a$. For $0 \le \tau \le 1$,

$$\frac{(1-\tau)^2}{(1+\lambda)^2(1+\omega)} \ge 1 - 2\lambda - 2\tau - \omega.$$

To see this without requiring the right side to be positive, use $1 - uvw \le (1-u) + (1-v) + (1-w)$ for $u, v, w \in [0,1]$, together with $1 - (1+\lambda)^{-2} \le 2\lambda$ and $1 - (1+\omega)^{-1} \le \omega$. Insert the tail and moment bounds into (20), and use $\rho_R a \ge a - a|\rho_R - 1|$. For the qualitative assertion choose $s = \sqrt{R}$. Then $s/R \to 0$, $T_f(s) \to 0$, and $B_f(R+s) \to 0$ by dominated convergence, while $\rho_R \to 1$. □

Under G and $I_1 < \infty$, F_ν satisfies the theorem's hypotheses. For nonzero ν, its a is positive: positivity of the Fourier transform gives $F_\nu(0) \ge 0$, and (18) has a strictly positive lattice sum. Thus the source's sharp lattice value is obtained directly from an actual mixed auxiliary, not just by integrating scalar energy inequalities. For inverse powers $p > 3/2$, $M_1 < \infty$ and one may take $m = 2p - 2$; for $1 < p \le 3/2$ use B_f rather than assert a nonexistent first moment. Schwartz mixtures permit any fixed m, with constants depending on the mixture and m. Nothing here gives a uniform rate for all completely monotone potentials or for the observed density deficit.

6.1 What perturbations are actually safe

A numerical bound $\widetilde{f}(x) \le \Phi(x) + \varepsilon$ gives only

$$\frac{1}{N}\sum_{i\neq j}\Phi(x_i - x_j) \ge \frac{1}{N}\sum_{i\neq j}\widetilde{f}(x_i - x_j) - \varepsilon(N-1).$$

The last term grows without bound at density one. Nor does a decaying error automatically solve the problem: an arbitrary number of points can be packed into a small region where the error is positive. Source G uses exact signs after controlled interpolation corrections, not an additive per-pair relaxation.

A valid alternative is relative error. If $\Phi \ge 0$, $\widetilde{f} \le (1+\varepsilon)\Phi$, $\widehat{\widetilde{f}} \ge 0$, and the other finite-window hypotheses hold, divide the lower bound for $\widetilde{f}$ by $1+\varepsilon$. This loses a multiplicative factor but requires no local count information. A spatially decaying additive error can be used only with a separate occupancy or energy-domination estimate. We prove no such estimate for all source competitors.

7 Comparison, costs, and research directions

The original TR147 inversion argument and the volume16 Bernstein proof serve different purposes here. The former enforces the missing gate between finite algebra and a realized infinite inverse; the latter explains why positive Bernstein coefficients certify a whole interval only after the nonpolynomial

remainder is bounded. Neither is credited as a new theorem. The local additions are the weighted bootstrap with the source's exact kernel exponent, the admissibility classification of its actual Gaussian family, and the quantitative disk transfer.

Public-primary checking on October 8, 2026 retained the following scope. Cohn–Kumar–Miller–Radchenko–Viazovska prove universal optimality in dimensions 8 and 24 and a radial interpolation theorem there [2]; this is not a planar uniqueness theorem. Sardari's Theorem 1.7 [3] gives infinitely many independent radial Schwartz functions with double vanishing of the function and its Fourier transform at triangular shell radii. We checked the theorem statement, not its entire proof. Our uniqueness is only in the source's fixed-list, fixed-extras, larger-node system. Gaussian subordination in [4] is established classical precedent, in a different extremal-function setting, not a substitute for the present measure bounds. Hardin–Tenpas [5] treat specified four-point and six-point periodic competitor families; Leble [6] proves local optimality among point configurations. The latter two primary abstracts and version metadata were checked, not their proofs. These results must not be described as proving the source's full centered-density theorem.

Advantages. The construction tolerates potentials singular at zero and measures of infinite total mass at large Gaussian parameter. Its endpoint conditions are both necessary and sufficient for this family. It exposes why a finite L^1 auxiliary need not be Schwartz, avoids an unjustified Poisson step for non-Schwartz mixtures, and retains arbitrary clustering in the finite-window estimate. The exponential coefficient norm additionally controls infinite-list truncation in a concrete source model.

Costs and limits. The family still depends on G: the cardinal representation, the exact infinite inverse and the uniform signs. An executed finite checker does not prove these reductions. Seminorm constants C_p are proved finite but not optimized numerically. The weighted bound degenerates near q_*, and admissibility excludes constant potentials and slow inverse-power tails from ordinary integrable LP theory. No minimum-configuration classification, general Fourier interpolation formula, renormalized screening theorem, or companion atomic certificate is deduced.

Feasible extensions. One can derive computable seminorm constants from the two compact complex Gaussian parameter sets, then optimize padding using measured T_f rather than a power majorant. Keeping the endpoint vanishing of the spectral density may improve weighted truncation constants; it does not follow that the limiting exponent is attainable. A useful further project is to certify an actual mixed auxiliary for one specified singular potential, with explicit inverse-moment and spatial-tail budgets. Such work should not silently replace the source's arbitrary-configuration class by separated or periodic competitors.

8 Verification and evidence boundaries

The owned read ledger distinguishes complete source-text reading from a global proof audit. Fresh private reading comprises TR147 lines 1770–1915 and the complete 299-line clearance chapter; the rest of TR147 and the private corpus were not inspected. No private text or title was submitted to a web service. Public searches used only public titles and identifiers. The pinned-source and captured-October-8 identity record supplied by the parent is provenance, not a proof or a live-branch assertion.

The source's original checkers were downloaded from the pinned commit into an owned audit directory, read completely, and kept unchanged. Their manifest checks 13 formula files and four checker/dependency files; it explicitly certifies integrity, not formula-to-code equivalence. A fresh

owned virtual environment uses the required python-flint 0.9.0. Original rerun outcomes and independently authored rational fixtures are recorded separately in the final evidence. Neither kind of test proves convergence of an infinite operator, continuum positivity by itself, or a screening argument. The unexecuted floor-rounded alternative remains explicitly unexecuted.

Both original commands exited with status zero. The 256-bit numeric run completed all 37,310 Bernstein comparisons and ten tail comparisons in approximately 340 seconds, as well as its power and column-norm tests. Its diagnostic minimum was enclosed near .00120611264; every comparison, not just this selected minimum, was tested. The 512-bit scalar run reported success and 107 named numerical records (not a claim of 107 total internal assertions). The programs rechecked their input hashes before writing the success records. The full floor-matrix computation and degree-2000 rational alternative were not run.

The separate standard-library Fraction script passed 4,918 exact fixtures: 402 kernel endpoint/grid fixtures, 528 periodic-node sums, 122 residual fixtures, 720 deleted-jet/degree-40 Bernstein fixtures, 385 Gaussian-moment recurrences, 2,601 disk error-factor inequalities, and 160 power-tail scalings. Each fixture counts once even when it asserts several identities. The polynomial fixtures are deliberately manufactured and do not substitute for the original coefficient arrays. Parent checks are not included in either author-run count. The proof ledger links each local result to its actual hypotheses; the freeze manifest binds the final source, PDF, tests, reading records, and all-page visual evidence.

References

[1] OpenAI, *Universal optimality of the triangular lattice*, September 23, 2026. Pinned public source at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, manuscript and verification directory.

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[3] N. Talebizadeh Sardari, *Higher Fourier interpolation on the plane*, 2021, Theorem 1.7. arXiv:2102.08753v2.

[4] E. Carneiro, F. Littmann, J. D. Vaaler, *Gaussian subordination for the Beurling–Selberg extremal problem*, Trans. Amer. Math. Soc. 365 (2013), 3493–3534. Primary preprint.

[5] D. Hardin, N. Tenpas, *Universally optimal periodic configurations in the plane*, Discrete Analysis (2025), article 26. Primary version 4.

[6] T. Leble, *The hexagonal lattice is universally locally optimal*, 2025. arXiv:2511.03353.

[7] M. Subbarao, confidential original working corpus: TR147, lines 1770–1915 (infinite analytic inversion and its domain gates); volume16, clearance chapter, lines 1–299 (Bernstein hull, subdivision, and separate remainder control). Exact local paths and hashes are retained in the owned read ledger; no public URL is supplied.